

Hospital Infection Analysis Report

The boxplots showed clear differences between patients with and without bloodstream infection. Patients with a bloodstream infection had a higher median age (71 vs 51 years) and a longer median hospital stay (17 vs 6 days). The infection group also had higher median WBC counts (17.1 vs 7.5) and CRP levels (220 vs 12). Overall, these findings suggest that older age, longer hospital stays, and higher levels of inflammatory markers, particularly CRP, and WBC were associated with bloodstream infection in this dataset.

The difference in age was also supported by a Welch's independent samples t-test. Patients with bloodstream infection had a higher mean age than those without infection (72.2 vs. 49.8 years), giving a mean difference of 22.4 years. This difference was statistically significant, $t(26.42) = -6.82$, $p = 2.798 \times 10^{-7}$ ($p < 0.001$).

A Pearson's chi-squared test was also conducted to assess the association between catheter use and bloodstream infection. A statistically significant association was found, $\chi^2(1) = 26.13$, $p = 3.186 \times 10^{-7}$ ($p < 0.001$).

Based on these findings, hospital infection control efforts could place particular attention on catheter management. This could include ensuring strict adherence to aseptic techniques during catheter insertion, using catheters only when clinically necessary, and providing regular refresher training for medical staff on catheter insertion, maintenance, and removal.

Older patients and patients with elevated WBC and CRP levels could also be monitored more closely for early signs of infection. In addition, reducing avoidable delays in treatment and improving discharge planning may help prevent unnecessarily prolonged hospital stays.

However, it is important to recognise that these results demonstrate associations rather than causation. The findings do not establish that factors such as catheter use, age, or length of hospital stay directly cause bloodstream infection. Further analysis would be useful to account for potential confounding factors, including age, underlying illness, hospital ward, severity of illness, and duration of catheterisation.

Finally, the sample size was relatively small ($n = 30$). A larger sample would provide more reliable estimates and reduce the influence that individual patients or unusual anomalies may have on the results. Future analysis using a larger dataset and multivariable statistical methods, such as logistic regression, could provide a more detailed and realistic assessment of factors associated with bloodstream infection.